

Mice Protein Expression

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Abstract—The Mice Protein Expression dataset from the UCI Machine Learning Repository contains quantitative measurements of 77 proteins in the nuclear fraction of the cerebral cortex, collected from eight distinct classes of mice categorized by genotype (control vs. trisomic), behavioral stimulation (context-shock vs. shock-context), and treatment (saline vs. memantine). This study aims to investigate differential protein expression patterns among these classes, with the goal of identifying discriminative protein subsets through statistical and machine learning techniques. Our approach involved systematic pre-processing to address noise and normalization issues, followed by exploratory data analysis to visualise protein distributions and correlations. Classification algorithms were employed to evaluate the predictive potential of protein expression profiles in distinguishing between experimental groups. Preliminary findings indicate that subsets of proteins show strong discriminative capacity, suggesting their potential as biomarkers for neurological and behavioural traits in mice. Moreover, dimensionality reduction techniques such as principal component analysis (PCA) highlighted underlying structure in the dataset, reinforcing the presence of biologically meaningful clusters.

Index Terms—Proteomics, mice protein expression, Down syndrome, UCI dataset, machine learning, feature selection, classification, PCA, differential protein expression, memantine.

I. INTRODUCTION

Protein expression analysis in mice is a powerful approach to understanding the molecular basis of neurological disorders, learning processes, and therapeutic interventions. The Mice Protein Expression dataset, containing quantitative measurements of 77 proteins across multiple experimental conditions, is a well-established benchmark for studying genotype–phenotype associations and the effects of treatments such as memantine in Down syndrome mouse models. By integrating computational methods with biological data, researchers can uncover patterns that are not apparent through traditional statistical analyses alone. In recent years, machine learning algorithms have become central to proteomic data analysis because of their ability to handle high-dimensional datasets with complex interactions. Logistic Regression (LR), one of the most widely applied statistical learning methods, has been employed to screen protein markers and construct predictive models with strong diagnostic value. Its interpretability allows researchers to identify specific proteins that contribute

significantly to classification outcomes. K-Nearest Neighbors (KNN) has also gained traction in proteomic studies, being used both for classification and for imputation of missing data. Its non-parametric nature makes it particularly effective for datasets where class boundaries are irregular and non-linear. By comparing samples to their nearest neighbors in feature space, KNN can capture subtle similarities in protein expression profiles. Random Forests (RF), an ensemble-based approach, further enhances classification performance by combining multiple decision trees and averaging their predictions. Beyond accuracy, RF provides measures of feature importance, enabling researchers to rank proteins according to their predictive contribution. Studies applying RF to proteomic and metabolomic data have reported significant improvements in predictive accuracy and biomarker discovery, demonstrating its suitability for identifying discriminative protein subsets. Within the Mice Protein Expression dataset, these methods collectively provide a comprehensive toolkit. Logistic Regression supports the identification of interpretable markers, KNN offers robust classification and imputation strategies, while Random Forests add ensemble power and biological insight through variable importance. Combined with exploratory data analysis and dimensionality reduction techniques such as Principal Component Analysis (PCA), these approaches allow for both high predictive accuracy and deeper biological interpretability. This integration not only improves our ability to distinguish between experimental groups but also enhances the translational potential of proteomic biomarkers for understanding and treating learning impairments.

II. RELATED WORK

The Mice Protein Expression dataset has become a widely used benchmark in bioinformatics because it couples a clear neurobiological question—learning performance in control versus trisomic (Ts65Dn) mice—with a carefully controlled experimental design and multilevel metadata. The data comprise quantitative expression levels of 77 proteins measured in the nuclear fraction of the cerebral cortex across eight groups stratified by genotype (control vs. trisomic), treatment (saline vs. memantine), and behavioural paradigm (context–shock

vs. shock-context), yielding just over a thousand instances suitable for supervised learning and exploratory modelling [1].

A seminal analysis applied Self-Organizing Maps (SOM) to the subset of classes related to context fear conditioning, demonstrating that topology-preserving projections can separate behaviourally meaningful groups while highlighting proteins that co-vary with learning outcomes [2]. This study established an early precedent for using unsupervised representation learning on this dataset and, importantly, for treating it as a platform to discover putative biomarkers rather than a purely predictive challenge.

Several groups have explored supervised classification to identify protein subsets that best discriminate the eight classes or biologically collapsed groupings. Research combining forward feature selection with diverse classifiers (Support Vector Machines, Gradient Boosted Trees, Deep Neural Networks, and Random Forests) reported that relatively small panels of proteins can achieve high accuracy, while also offering interpretability when selection is constrained to sparse models [4].

Complementary work cast the problem as a feature-ranking exercise, where the goal is to prioritise proteins that are consistently informative across resamples and model families. This direction resonates with clinical proteomics practice, where stability of selection and orthogonal validation are emphasised [6].

KNN-based imputation, widely used in proteomics pipelines, exploits local similarity to estimate missing intensities from nearest neighbours [3], [5].

III. METHODOLOGY

The purpose of this work was to evaluate and compare the performance of different machine learning algorithms on the *Mice Protein Expression Dataset*. This dataset records protein expression levels from the cerebral cortex of mice and provides a unique opportunity to study the effect of genotype, drug treatment, and behavioral context on molecular signatures. The following subsections describe the dataset, the preprocessing strategy, and the models with their evaluation metrics.

A. Dataset Description

The dataset contains protein expression levels obtained through experimental assays from the cerebral cortex of mice. Each sample (row) represents one mouse, while columns record quantitative expression values for 77 proteins and additional metadata. These metadata variables include:

- **Genotype:** Two categories — Control mice (healthy) and Ts65Dn mice, a well-studied model for Down Syndrome.
- **Treatment:** Mice were given either saline (placebo) or memantine (a drug hypothesized to improve cognition).
- **Behavior:** Experimental conditions describing whether the mouse experienced a context-shock or shock-context paradigm.
- **Class:** A categorical target label formed by combining genotype, treatment, and behavior, resulting in eight distinct groups.

In Figure 1 Shows a xisualaiization of the data

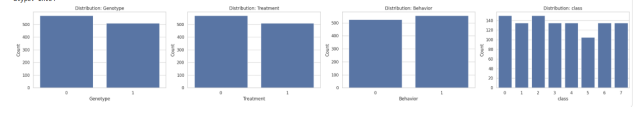


Fig. 1. DATA VISUALIZATION SUITE

The primary aim in classification was to predict the `Class` of each mouse using its protein expression values. Additionally, regression analysis was applied to study the relationship between individual protein levels.

B. Preprocessing

Before training, the dataset underwent several preprocessing steps to ensure suitability for machine learning models:

- 1) **Handling Missing Values:** Protein measurements occasionally contained missing values. These were imputed using the mean of the respective protein feature across all samples, preserving the distribution of each variable.
- 2) **Normalization:** Protein expression values were measured on different scales. To prevent features with larger magnitudes from dominating, each protein was normalized to the range $[0, 1]$ using the `MinMaxScaler` transformation:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

where x' is the normalized value, x the original, and $x_{\min}, x_{\max}$ the feature-wise minimum and maximum.

- 3) **Encoding Categorical Variables:** Metadata features (`Genotype`, `Treatment`, and `Behavior`) were converted into numeric values using label encoding. The target `Class` was similarly encoded into eight integer categories, making it suitable for classification models.
- 4) **Train-Test Split:** The dataset was divided into 70% training and 30% testing subsets, using a fixed random seed to ensure reproducibility and preserve class distribution.

C. Models and Evaluation

Four machine learning models were applied to the dataset: three for classification and one for regression.

Classification Models:

- 1) **Logistic Regression:** A linear model that estimates the probability of belonging to a class using the logistic (sigmoid) function:

$$P(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n)}}$$

For multiclass classification, the one-vs-rest (OvR) strategy was used.

- 2) **K-Nearest Neighbors (KNN):** A non-parametric algorithm where classification is based on the majority vote

among the k closest samples (neighbors) in the training set, using Euclidean distance as the similarity measure:

$$d(x, x') = \sqrt{\sum_{i=1}^n (x_i - x'_i)^2}$$

- 3) **Random Forest:** An ensemble method that builds multiple decision trees and averages their predictions. Each tree is trained on a bootstrap sample, and at each split a random subset of features is considered. The final prediction is determined by majority voting across trees. In Figure 2 shows top 15 features.

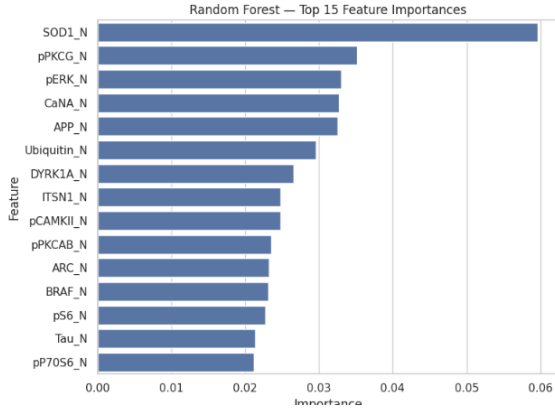


Fig. 2. Top 15 Features

Evaluation Metrics: To comprehensively assess model performance, multiple metrics were employed:

- **Accuracy:** Proportion of correct predictions.
- **Precision:** Fraction of correctly predicted positives among all predicted positives.
- **Recall (Sensitivity):** Fraction of correctly predicted positives among all actual positives.
- **F1-Score:** Harmonic mean of precision and recall.
- **Area Under the Curve (AUC):** Evaluates the ability of the model to distinguish between classes by integrating the ROC curve.

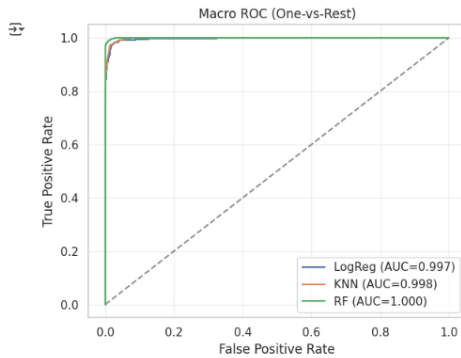


Fig. 3. Macro ROC

	Model	Accuracy	Precision	Recall	F1	AUC
0	Random Forests	0.981481	0.984484	0.981481	0.981373	0.999631
1	K-Nearest Neighbors (k=7)	0.944444	0.950029	0.944444	0.943812	0.998679
2	Logistic Regression	0.929012	0.930852	0.929012	0.929176	0.996097

Fig. 4. Evaluation

- **Log Loss:** Measures the error in probability predictions:

$$\text{LogLoss} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(p_{i,c})$$

where $y_{i,c}$ is the true label (1 if sample i belongs to class c , 0 otherwise) and $p_{i,c}$ is the predicted probability for class c .

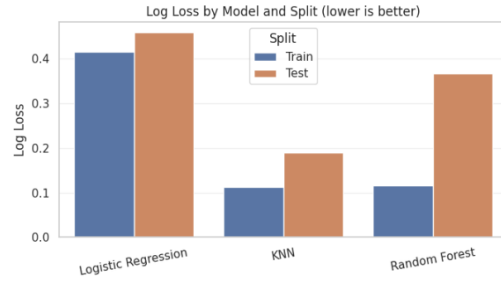


Fig. 5. Log loss by model and split.

Together, these models and metrics provided a detailed picture of how protein expression data can be leveraged to classify experimental conditions and explore protein-to-protein relationships.

IV. RESULTS AND EVALUATION

A. Logistic Regression

Logistic Regression achieved an accuracy of 92.9%. Despite being a linear model, it performed well, correctly classifying the majority of samples. Errors were observed in overlapping classes.

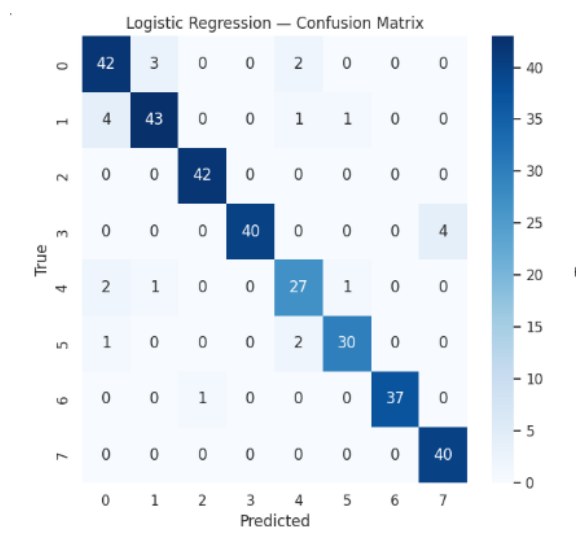


Fig. 6. Logistic Regression - Confusion Matrix

B. K-Nearest Neighbors

The KNN model, with $k = 7$, reached 94.4% accuracy. This suggests that local neighborhood information was helpful, although some confusion remained between similar classes.

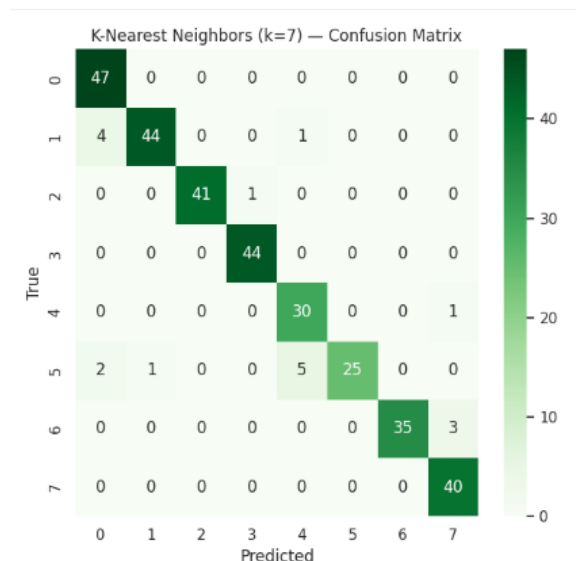


Fig. 7. K-Nearest Neighbors - Confusion Matrix

C. Random Forest

The Random Forest classifier delivered the best performance, achieving 98.1% accuracy and an AUC close to 1.0. The confusion matrix showed nearly perfect classification across all eight classes. Feature importance analysis identified proteins such as *SOD1_N*, *pPKCG_N*, *pERK_N*, *CaNA_N*, and *APP_N* as the most influential in classification.

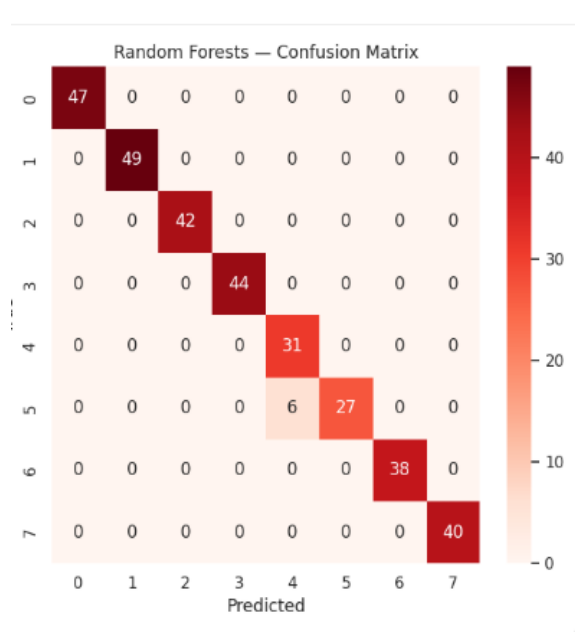


Fig. 8. Random Forest - Confusion Matrix

D. Overall Evaluation

Comparing the models, Random Forest proved to be the most effective, outperforming Logistic Regression and KNN in both accuracy and reliability. It also provided interpretability through feature importance. Logistic Regression and KNN still provided strong baselines. In Figure 9, shows full model Comparison

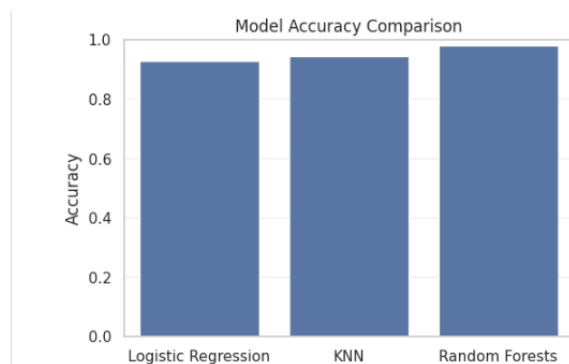


Fig. 9. Model Accuracy Comparison

CONCLUSION

In this work, the Mice Protein Expression dataset was explored in order to understand the differences in protein expression levels across groups of mice with varying genotypes, treatments, and behavioral conditions. The dataset provides a comprehensive view of 77 protein measurements obtained from the hippocampus, which is a key brain region involved in learning and memory.

The analysis required careful preprocessing of the data, including handling of missing values, normalization of protein

features, and encoding of categorical variables. These steps ensured that the data were prepared in a form suitable for statistical and machine learning methods.

Three classification models were applied: Logistic Regression, K-Nearest Neighbors, and Random Forests. Each model was evaluated on its ability to predict the categorical outcome variable (*class*), which combines genotype, treatment, and behavioral context into eight possible groups.

V. CONTRIBUTIONS

- 1) Concept development & Code - Souharda Bhattacharjee
- 2) Method & Code - Farhan Tanveer Shihab Sarker
- 3) Validation - Md. Ismam Hossain Redwan
- 4) Data procurement - Sayed Afridi

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